INCH-POUND

MS17825J
27 June 2011
SUPERSEDING
MS17825H
5 February 2004

DETAIL SPECIFICATION SHEET

# NUT, SELF-LOCKING, CASTELLATED, HEXAGON, 250 °F NON-METALLIC INSERT

Reactivate after 5 February 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and NASM25027.

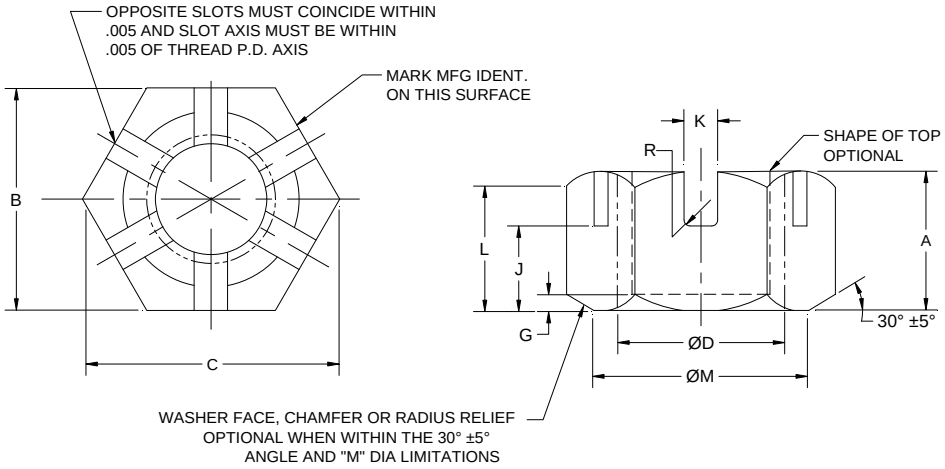


FIGURE 1. NUT.

TABLE I. DASH NUMBERS, DIMENSIONS AND STRENGTHS.

| SIZE DASH NO. | THREAD AS8879 | A | | B | | C | ∅D | G | | J | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAX | MIN | MAX | MIN | MIN | MAX | MAX | MIN | MAX | MIN |
| -3 | .1900-32 UNJF-3B | .265 | .235 | .376 | .367 | .410 | .215 | .032 | .006 | .125 | .105 |
| -4 | .2500-28 UNJF-3B | .296 | .266 | .439 | .430 | .482 | .275 | .036 | .007 | .141 | .121 |
| -5 | .3125-24 UNJF-3B | .343 | .313 | .502 | .492 | .552 | .338 | .042 | .008 | .188 | .168 |
| -6 | .3750-24 UNJF-3B | .421 | .391 | .564 | .553 | .622 | .400 | | | .235 | .215 |
| -7 | .4375-20 UNJF-3B | .468 | .438 | .690 | .679 | .766 | .463 | .050 | .010 | .282 | .262 |
| -8 | .5000-20 UNJF-3B | .578 | .548 | .752 | .741 | .837 | .525 | | | .375 | .355 |
| -9 | .5625-18 UNJF-3B | .624 | .594 | .877 | .865 | .978 | .587 | .055 | .012 | .406 | .386 |
| -10 | .6250-18 UNJF-3B | .734 | .704 | .940 | .928 | 1.051 | .650 | | | .484 | .464 |
| -12 | .7500-16 UNJF-3B | .828 | .798 | 1.064 | 1.052 | 1.191 | .795 | .063 | .014 | .578 | .558 |
| -14 | .8750-14 UNJF-3B | .921 | .891 | 1.252 | 1.239 | 1.403 | .900 | .072 | .016 | .672 | .652 |
| -16 | 1.0000-12 UNJF-3B | 1.015 | .985 | 1.440 | 1.427 | 1.615 | 1.025 | | | .766 | .746 |
| -18 | 1.1250-12 UNJF-3B | 1.171 | 1.141 | 1.627 | 1.614 | 1.825 | 1.150 | .083 | .018 | .828 | .808 |
| -20 | 1.2500-12 UNJF-3B | 1.265 | 1.235 | 1.815 | 1.801 | 2.038 | 1.275 | | | .891 | .871 |



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TABLE I. DASH NUMBERS, DIMENSIONS AND STRENGTHS.. (Continued)

| SIZE DASH NO. | K | | L | ∅M | | R | ULTIMATE TENSILE STRENGTH LB MIN | TEST WRENCHING TORQUE (IN. LBS) | ASSEMBLY TORQUE VIBRATION (IN. LBS) | WEIGHT LB/100 |
|---|---|---|---|---|---|---|---|---|---|---|
| | MAX | MIN | MIN | MAX | MIN | MAX | | MIN | MAX | MAX |
| -3 | .108 | .078 | .110 | .376 | .355 | .035 | 1,950 | 35 | 28 | .400 |
| -4 | | | .125 | .439 | .418 | | 3,500 | 80 | 65 | .650 |
| -5 | | | .172 | .502 | .480 | | 5,500 | 225 | 180 | 1.000 |
| -6 | .155 | .125 | .218 | .564 | .543 | .053 | 9,000 | 325 | 260 | 1.400 |
| -7 | | | .265 | .690 | .675 | | 12,200 | 575 | 460 | 2.000 |
| -8 | | | .359 | .752 | .737 | | 16,600 | 900 | 720 | 3.900 |
| -9 | .187 | .157 | .390 | .877 | .862 | .062 | 21,200 | 1,100 | 880 | 6.900 |
| -10 | | | .468 | .940 | .925 | | 27,400 | 1,600 | 1,300 | 11.400 |
| -12 | | | .562 | 1.064 | 1.049 | | 40,000 | 2,800 | 2,200 | 16.900 |
| -14 | | | .656 | 1.252 | 1.237 | | 54,600 | 4,600 | 3,700 | 26.000 |
| -16 | | | .750 | 1.440 | 1.425 | | 71,000 | 6,800 | 5,400 | 38.000 |
| -18 | | | .844 | 1.627 | 1.612 | | 92,600 | 10,000 | 8,000 | 53.500 |
| -20 | | | .938 | 1.815 | 1.800 | | 118,500 | 14,000 | 11,000 | 74.500 |

REQUIREMENTS:

1. MATERIALS: Steel, UNS G11370 (AISI 1137) per AMS5024
   UNS G41300 (AISI 4130) per AMS6348
   UNS G87400 (AISI 8740) per AMS6322, 6323, 6325, 6327, and 6358
   INSERT MATERIAL: Nylon.
2. FINISH: Cadmium plating in accordance with SAE-AMS-QQ-P-416, Type II, Class 2.
3. Break all sharp edges and remove burrs.
4. SURFACE TEXTURE: 125√ in accordance with ASME B46.1.
5. DIMENSIONS: All dimensions are in inches unless otherwise specified.
6. DESIGN INFORMATION: These nuts may be used at temperatures 250° F and below in those types of applications for which slotted or castellated nuts are approved for use, and if such applications conflict with any limitation on the use of self-locking nuts, as cited by NASM33588, this standard takes precedence. Nuts are to be installed in accordance with NASM33540.
7. EXAMPLE OF PART NUMBER: MS17825-4 = .2500-28 nut, cadmium plated, non-metallic insert.
8. PERFORMANCE: Per NASM25027 except for:
   1 – Tabulated ultimate tensile strength values.
   2 – Vibration per NASM1312-7, (30,000 cycles) using assembly torques listed in Table I for one seated cycle.
   3 – Locking torque valves applicable for one cycle at room temperature only.
9. For design feature purposes, this standard takes precedence over procurement document referenced herein.
10. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 99

Preparing activity:
DLA-IS

(Project 5310-2011-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil .



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